Calculus and Linear Algebra

Complete Function Study

Oblique asymptotes and remarkable limits

Lecture 23

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What is a complete function study?

A complete study organizes all the tools developed so far into one coherent analysis.

1. Domain.
2. Symmetries and intersections with the axes.
3. Sign of the function.
4. Limits and asymptotes.
5. First derivative: monotonicity and extrema.
6. Second derivative: concavity and inflection points.
7. Final graph.

The order matters

A good function study is not a collection of unrelated computations.

Recommended workflow

$$D \to \text{symmetry} \to \text{zeros/sign} \to \text{limits} \to f' \to f'' \to \text{graph}$$

Each step restricts or confirms what the next steps can look like.

Before differentiating: read the formula

Ask first:

- Is the function rational, logarithmic, exponential, radical, trigonometric?
- Where can it be undefined?
- Is it even or odd?
- Are obvious zeros or intercepts available?
- Is there an evident asymptotic behaviour?

Efficiency

Do not compute derivatives before understanding the basic geometry imposed by the formula.

Example 1: a complete rational study

We study

$$\boxed{f(x) = \frac{x^2}{x-1}.}$$

This example contains:

- a domain restriction;
- a vertical asymptote;
- an oblique asymptote;
- stationary points;
- concavity on disconnected intervals.

Step 1: domain

The denominator must be non-zero:

$$x - 1 \neq 0.$$

Hence

$$\boxed{D = \mathbb{R} \setminus \{1\}.}$$

The point $x = 1$ splits the analysis into two independent intervals:

$$(-\infty, 1), \qquad (1, +\infty).$$

Step 2: symmetry and intersections

The function is neither even nor odd. For intersections with the x-axis:

$$f(x) = 0 \iff x^2 = 0 \iff x = 0.$$

Since

$$f(0) = 0,$$

the graph intersects both axes at

$$\boxed{(0, 0).}$$

The zero has multiplicity 2, so the sign need not change there.

Step 3: sign

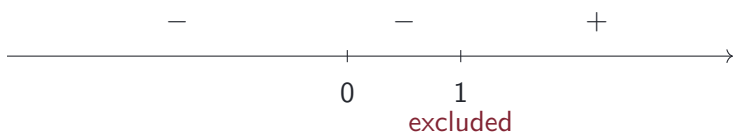
Because $x^2 \geq 0$, the sign is determined by $x - 1$ except at $x = 0$.

$$f(x) < 0 \quad \text{for } x < 1,\ x \neq 0,$$

$$f(x) > 0 \quad \text{for } x > 1,$$

with

$$f(0) = 0.$$



Step 4: limit near the excluded point

At $x = 1$,

$$x^2 \to 1 > 0.$$

Therefore

$$\lim_{x \to 1^-} \frac{x^2}{x-1} = -\infty, \qquad \lim_{x \to 1^+} \frac{x^2}{x-1} = +\infty.$$

Hence

$$\boxed{x = 1 \text{ is a vertical asymptote.}}$$

Step 4: behaviour at infinity

Polynomial division gives

$$\frac{x^2}{x-1} = x + 1 + \frac{1}{x-1}.$$

Thus

$$f(x) - (x+1) = \frac{1}{x-1} \to 0 \qquad (x \to \pm\infty).$$

Therefore

$$\boxed{y = x + 1 \text{ is an oblique asymptote.}}$$

Why polynomial division is useful

The decomposition

$$f(x) = x + 1 + \frac{1}{x - 1}$$

immediately separates:

- the dominant linear behaviour $x + 1$;
- the small correction $1/(x - 1)$;
- the vertical singularity at $x = 1$.

General principle

For rational functions with numerator degree exactly one larger than denominator degree, look for an oblique asymptote.

Oblique asymptotes: the general definition

A line

$$y = mx + q$$

is an **oblique asymptote** of f as $x \to +\infty$ if

$$\boxed{\lim_{x \to +\infty} \big[f(x) - (mx + q)\big] = 0.}$$

The same definition applies as $x \to -\infty$.

Geometric meaning

Far from the origin, the vertical distance between the graph and the line tends to zero.

How to compute an oblique asymptote

Suppose an oblique asymptote $y = mx + q$ exists as $x \to +\infty$. Then

$$\boxed{m = \lim_{x \to +\infty} \frac{f(x)}{x}}$$

provided this limit is finite and non-zero. Once m is known,

$$\boxed{q = \lim_{x \to +\infty} \big[f(x) - mx\big].}$$

Important

If $m = 0$, the asymptote is horizontal rather than oblique.

Example: compute m and q directly

Consider

$$f(x) = \frac{2x^2 + 3}{x - 1}.$$

First,

$$m = \lim_{x \to +\infty} \frac{f(x)}{x} = \lim_{x \to +\infty} \frac{2x^2 + 3}{x(x - 1)} = 2.$$

Then

$$q = \lim_{x \to +\infty} \left(\frac{2x^2 + 3}{x - 1} - 2x \right) = \lim_{x \to +\infty} \frac{2x + 3}{x - 1} = 2.$$

Hence

$$\boxed{y = 2x + 2.}$$

Rational functions: degree and asymptotes

Let $f(x) = P(x)/Q(x)$ with polynomials P, Q.

Degree relation	Behaviour at infinity
$\deg P < \deg Q$	$y = 0$ is a horizontal asymptote
$\deg P = \deg Q$	horizontal asymptote = ratio of leading coefficients
$\deg P = \deg Q + 1$	look for an oblique asymptote
$\deg P > \deg Q + 1$	polynomial asymptotic behaviour of higher degree

For rational functions

Polynomial division is usually the fastest way to reveal the asymptotic polynomial.

Step 5: first derivative

Using the decomposed form,

$$f'(x) = 1 - \frac{1}{(x-1)^2}.$$

Equivalently,

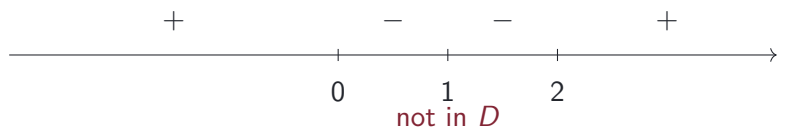
$$f'(x) = \frac{(x-1)^2 - 1}{(x-1)^2} = \frac{x(x-2)}{(x-1)^2}.$$

Critical points in the domain:

$$\boxed{x = 0, \qquad x = 2.}$$

Step 5: derivative sign chart

Since $(x-1)^2 > 0$ on the domain, the sign of f' is determined by $x(x-2)$.



Thus f increases on $(-\infty, 0)$ and $(2, +\infty)$, and decreases on $(0, 1)$ and $(1, 2)$.

Step 5: local extrema

At $x = 0$,

$$f' : + \to -,$$

so $(0, 0)$ is a local maximum. At $x = 2$,

$$f' : - \to +,$$

so

$$f(2) = 4,$$

and

$$\boxed{(2, 4) \text{ is a local minimum.}}$$

Step 6: second derivative

From

$$f'(x) = 1 - (x-1)^{-2},$$

we obtain

$$f''(x) = \frac{2}{(x-1)^3}.$$

Hence

$$f''(x) < 0 \quad \text{for } x < 1,$$

$$f''(x) > 0 \quad \text{for } x > 1.$$

The graph is concave on $(-\infty, 1)$ and convex on $(1, +\infty)$.

Is $x = 1$ an inflection point?

No. Although the sign of f'' changes across 1, the function is not defined there.

Therefore there is no point

$$(1, f(1))$$

on the graph.

$$\boxed{x = 1 \text{ is not an inflection point.}}$$

Important

A change of sign of f'' across a hole or asymptote does not automatically create an inflection point.

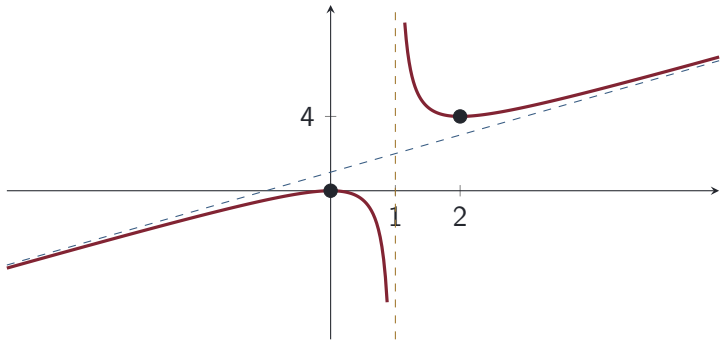
Complete qualitative summary

For $f(x) = x^2/(x-1)$:

- domain: $\mathbb{R} \setminus \{1\}$;
- zero: $(0, 0)$;
- vertical asymptote: $x = 1$;
- oblique asymptote: $y = x + 1$;
- local maximum: $(0, 0)$;
- local minimum: $(2, 4)$;
- concave for $x < 1$, convex for $x > 1$;
- no inflection point.

Now the graph is almost forced.

Final graph



Every visible feature corresponds to a result obtained analytically.

Example 2: a polynomial study

Now study

$$\boxed{g(x) = x^3 - 3x.}$$

Because it is a polynomial:

$$D = \mathbb{R}.$$

Moreover,

$$g(-x) = -g(x),$$

so g is odd and its graph is symmetric with respect to the origin.

Example 2: zeros and sign

$$g(x) = x(x^2 - 3) = x(x - \sqrt{3})(x + \sqrt{3}).$$

Zeros:

$$-\sqrt{3}, \quad 0, \quad \sqrt{3}.$$

The sign alternates across these simple zeros:

$$-, \ +, \ -, \ +.$$

There are no finite-domain singularities or vertical asymptotes.

Example 2: first derivative

$$g'(x) = 3x^2 - 3 = 3(x-1)(x+1).$$

Thus

$$g' > 0 \text{ on } (-\infty, -1) \cup (1, +\infty),$$

$$g' < 0 \text{ on } (-1, 1).$$

Hence

$(-1, 2)$ local maximum, $(1, -2)$ local minimum.

Example 2: second derivative

$$g''(x) = 6x.$$

Therefore

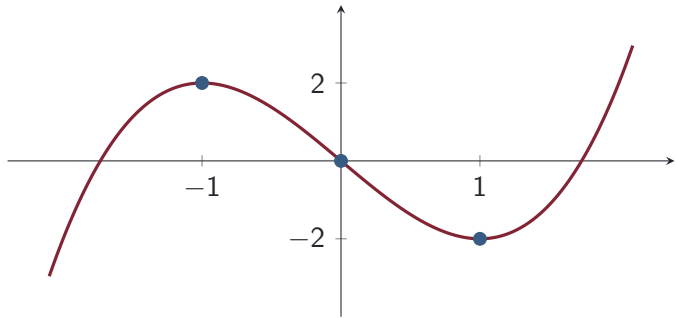
$$g'' < 0 \text{ for } x < 0, \qquad g'' > 0 \text{ for } x > 0.$$

The graph changes concavity at

$$\boxed{(0,0)},$$

which is an inflection point.

Example 2: final graph



The symmetry visible in the graph was already known before differentiating.

Remarkable limits: a first reference table

Several limits recur so often that they become standard tools.

$$\lim_{x\to 0}\frac{\sin x}{x}=1 \qquad \lim_{x\to 0}\frac{1-\cos x}{x^2}=\frac{1}{2}$$

$$\lim_{x\to 0}\frac{e^x-1}{x}=1 \qquad \lim_{x\to 0}\frac{\ln(1+x)}{x}=1$$

$$\lim_{x\to 0}(1+x)^{1/x}=e.$$

Today

The aim is to recognize and use them. Their deeper connections will reappear with Taylor expansions.

The sine limit with a generic constant

Starting from

$$\lim_{u \to 0} \frac{\sin u}{u} = 1,$$

set $u = ax$, with $a \in \mathbb{R}$ fixed. For $a \neq 0$,

$$\frac{\sin(ax)}{x} = a \, \frac{\sin(ax)}{ax}.$$

Hence

$$\boxed{\lim_{x \to 0} \frac{\sin(ax)}{x} = a.}$$

Equivalently,

$$\boxed{\lim_{x \to 0} \frac{\sin(ax)}{ax} = 1.}$$

The case $a = 0$ is immediate.

A useful consequence: cosine

Using

$$1 - \cos u = 2 \sin^2 \left(\frac{u}{2} \right),$$

we obtain

$$\lim_{u \to 0} \frac{1 - \cos u}{u^2} = \frac{1}{2}.$$

With $u = ax$,

$$\frac{1 - \cos(ax)}{x^2} = a^2 \frac{1 - \cos(ax)}{(ax)^2}.$$

Therefore

$$\boxed{\lim_{x \to 0} \frac{1 - \cos(ax)}{x^2} = \frac{a^2}{2}.}$$

The exponential limit with a generic constant

The basic limit is

$$\lim_{u \to 0} \frac{e^u - 1}{u} = 1.$$

Setting $u = ax$ gives

$$\frac{e^{ax} - 1}{x} = a \, \frac{e^{ax} - 1}{ax}.$$

Thus, for every fixed $a \in \mathbb{R}$,

$$\boxed{\lim_{x \to 0} \frac{e^{ax} - 1}{x} = a.}$$

For example,

$$\lim_{x \to 0} \frac{e^{3x} - 1}{x} = 3.$$

The logarithmic limit with a generic constant

From

$$\lim_{u\to 0} \frac{\ln(1+u)}{u} = 1,$$

set $u = ax$:

$$\frac{\ln(1+ax)}{x} = a\,\frac{\ln(1+ax)}{ax}.$$

Therefore

$$\boxed{\lim_{x\to 0} \frac{\ln(1+ax)}{x} = a.}$$

Of course, $1 + ax > 0$ near 0 whenever x is sufficiently close to 0.

The power limit with a generic exponent

For a fixed real number a,

$$\boxed{\lim_{x \to 0} \frac{(1+x)^a - 1}{x} = a.}$$

More generally, replacing x by bx,

$$\boxed{\lim_{x \to 0} \frac{(1+bx)^a - 1}{x} = ab.}$$

This is a useful bridge between remarkable limits and derivatives, because it is exactly the local behaviour of the power function near 1.

The limit that defines e

A fundamental exponential limit is

$$\boxed{\lim_{x\to 0}(1+x)^{1/x} = e.}$$

Equivalent form:

$$\boxed{\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^{n} = e.}$$

More generally, for a fixed $a \in \mathbb{R}$,

$$\boxed{\lim_{x\to 0}(1+ax)^{1/x} = e^{a}.}$$

Indeed,

$$(1+ax)^{1/x} = \left[(1+ax)^{1/(ax)}\right]^{a}.$$

Recognizing the pattern

Exercise Compute the following:

$$\text{A: } \lim_{x\to 0} \frac{\sin(4x)}{x}, \qquad \text{B: } \lim_{x\to 0} \frac{e^{-2x}-1}{x},$$

$$\text{C: } \lim_{x\to 0} \frac{\ln(1+5x)}{x}, \qquad \text{D: } \lim_{x\to 0} \frac{1-\cos(3x)}{x^2}.$$

Strategy

Rewrite each expression until one of the standard forms becomes visible.

Solutions: pattern recognition

Using the generalized remarkable limits,

$$\lim_{x\to 0} \frac{\sin(4x)}{x} = 4,$$

$$\lim_{x\to 0} \frac{e^{-2x}-1}{x} = -2,$$

$$\lim_{x\to 0} \frac{\ln(1+5x)}{x} = 5,$$

$$\lim_{x\to 0} \frac{1-\cos(3x)}{x^2} = \frac{9}{2}.$$

Main point

The constants are not new formulas to memorize: they come from a simple substitution in the basic remarkable limits.

A practical function-study checklist

Step	Question
Domain	Where does the function exist?
Symmetry	Even, odd, periodic, none?
Intersections/sign	Where is $f = 0$, $f > 0$, $f < 0$?
Limits	What happens near boundaries and at infinity?
Asymptotes	Vertical, horizontal, oblique?
f'	Where does the graph rise/fall? Extrema?
f''	Where does it bend? Inflection points?
Graph	Is every feature consistent?

Do not overcompute

Not every function requires every possible step.

- A polynomial has no finite-domain restrictions.
- A rational function may require asymptotes.
- Symmetry can halve the work.
- Some functions have no zeros.
- Sometimes f'' is unnecessary for a specific question.

Goal

Choose the calculations that reveal the graph, rather than following a rigid ritual.

Exercise 1: full study

Exercise Study completely

$$f(x) = \frac{x+1}{x-1}.$$

Determine:

- domain, zeros and sign;
- all asymptotes;
- monotonicity and extrema;
- concavity and inflection points;
- final sketch.

Exercise 1: solution, basic features

$$D = \mathbb{R} \setminus \{1\}, \qquad f(x) = 1 + \frac{2}{x-1}.$$

Zero:

$$x = -1.$$

Asymptotes:

$$\boxed{x = 1}, \qquad \boxed{y = 1}.$$

The sign is positive on $(-\infty, -1) \cup (1, +\infty)$ and negative on $(-1, 1)$.

Exercise 1: solution, derivatives

$$f'(x) = -\frac{2}{(x-1)^2} < 0$$

on both domain intervals, so f is strictly decreasing and has no extrema.

$$f''(x) = \frac{4}{(x-1)^3}.$$

Thus f is concave for $x < 1$ and convex for $x > 1$, but has no inflection point because $x = 1$ is outside the domain.

Exercise 2: full study

Exercise Study completely

$$g(x) = xe^{-x}.$$

Useful facts:

$$e^{-x} > 0 \quad \forall x,$$

$$g'(x) = e^{-x}(1 - x).$$

Do not start from the derivative: first determine domain, sign, zeros and limits.

Exercise 2: solution, first half

$$D = \mathbb{R}, \qquad g(x) = 0 \iff x = 0.$$

Since $e^{-x} > 0$, g has the sign of x.

$$\lim_{x \to +\infty} xe^{-x} = 0, \qquad \lim_{x \to -\infty} xe^{-x} = -\infty.$$

Thus

$$y = 0$$

is a horizontal asymptote as $x \to +\infty$ only.

Exercise 2: solution, derivatives

$$g'(x) = e^{-x}(1 - x).$$

Hence g increases on $(-\infty, 1)$ and decreases on $(1, +\infty)$, with maximum

$$\boxed{(1, e^{-1})}.$$

Also

$$g''(x) = e^{-x}(x - 2).$$

So g is concave on $(-\infty, 2)$ and convex on $(2, +\infty)$, with inflection point

$$\boxed{(2, 2e^{-2})}.$$

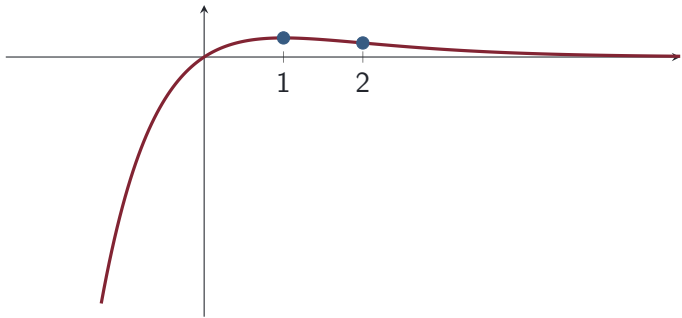
Graph prediction before plotting

For $g(x) = xe^{-x}$ we already know:

- starts from $-\infty$ as $x \to -\infty$;
- crosses the origin;
- reaches one maximum at $x = 1$;
- changes concavity at $x = 2$;
- approaches 0 from above as $x \to +\infty$.

The graph is now almost completely determined without software.

Exercise 2: final graph



The graph confirms every analytic conclusion.

Independent practice

Exercise Perform a complete study of one of the following:

$$\text{A: } f(x) = \frac{x^2+1}{x},$$

$$\text{B: } g(x) = \frac{x^2}{x^2-1},$$

$$\text{C: } h(x) = x^2 e^{-x}.$$

Produce a final graph only after completing the analytic table.

What a final answer should contain

A complete study should not end with a list of derivatives. It should clearly state:

- the domain and all exceptional points;
- intercepts and sign;
- asymptotes with the relevant limits;
- monotonicity intervals and extrema;
- concavity intervals and inflection points;
- a coherent final sketch.

Looking ahead

We now have the full machinery for qualitative graph analysis. Next steps will refine the calculus toolbox with topics such as:

- systematic use of remarkable limits and L'Hôpital's rule;
- Taylor approximation;
- integration;
- later, functions of two variables.

Core achievement

A formula can now be translated systematically into the geometry of its graph.